Profile/Power10 Performance Summary

FIS Profile V7.6.7 / GT.M V7.1

IBM Power10 – AIX 7.2

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Empowering
the Financial World

Performance Testing Objectives

- **Preparation for BIDV Bank, Vietnam, Go Live Launch on Power10**
- **Profile V7.6.7 / GT.M V7.1 Performance Analysis with IBM**
  - Reestablish IBM Power Series/ AIX Architecture Knowledgebase
  - Development of Profile/POWER10 Deployment Checklist
  - Deploy Storage and JFS2 Best Practices
  - IBM/GT.M Detailed Performance Analysis
    - Buildout 25M Customer/Account Database
    - Testing Includes Online Simulation and Key Overnight Batch Jobs
    - Encrypted Database
    - Spanned Global Regions
    - File System CIO – Global Buffer Testing
    - GTM Iterative Build Comparisons
    - SMT settings
  - Produce Final Performance Summary Findings

Power10/AIX 7.2 Configuration

- **IBM 9080-HEX POWER10 Server**
  - 15 X 3.550 GHz PowerPC_POWER10 64 Bit Processor (POWER9 Implementation Mode)
  - 150 GB Memory
  - AIX 7.2.5.103 TL05
- **Virtualized SAN Storage (FS9200) -10TB flash-based storage backend**
- **File System Layout**

JFS2 File System	Size (GB)
/profile	100
/profile_gbls1	300
/profile_gbls2	300
/profile_gbls3	300
/profile_gbls4	300
/profile_gbls5	300
/profile_gbls6	300
/profile_gbls7	300
/profile_gbls8	300
/profile_gbls9	300
/profile_gbls10	80
/profile_gbls11	80
/profile_gbls12	80
/profile_gbls13	80
/profile_gbls14	80

JFS2 File System	Size (GB)
/profile_journals1	240
/profile_journals2	240
/profile_journals3	240
/profile_journals4	240
/profile_journals5	80
/profile_journals6	80
/profile_journals7	80
/profile_journals8	80
/profile_journals9	80
/profile_spool	25
/profile_scratch	1000
/profile_backup	2000

Profile Configuration

- **Profile V7.6.7**
  - Profile master_v76_5589_26
  - Framework master_v40_5158_21246
- **25 Million Customer and Account Database**

Class	Type	Number of Customers
CIF	Personal	25,000,000

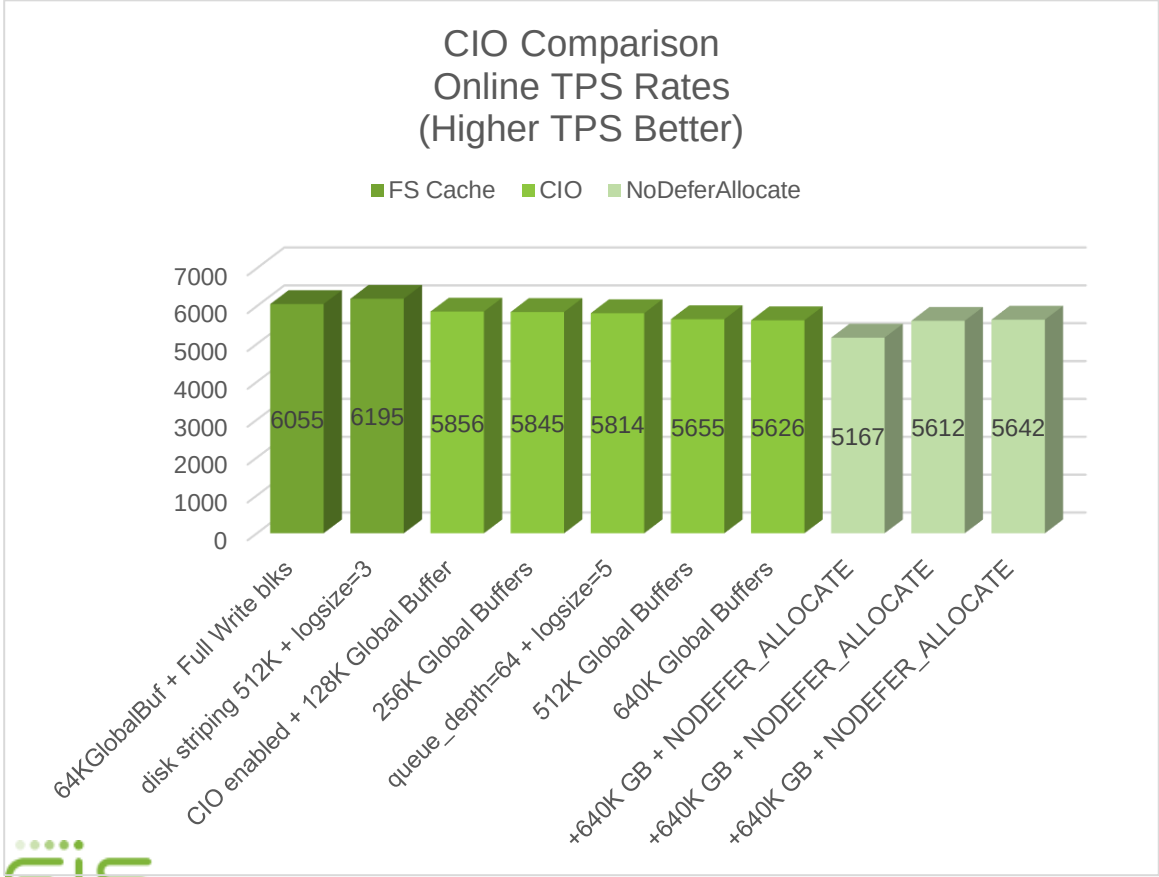
Class	Group	Type	Number of Accounts
Deposit	SAV	300	5,000,000
Deposit	CD	350	2,500,000
Deposit	DDA	400	10,000,000
Loan	LN	500	2,500,000
Loan	RC	600	2,500,000
Loan	MTG	700	2,500,000

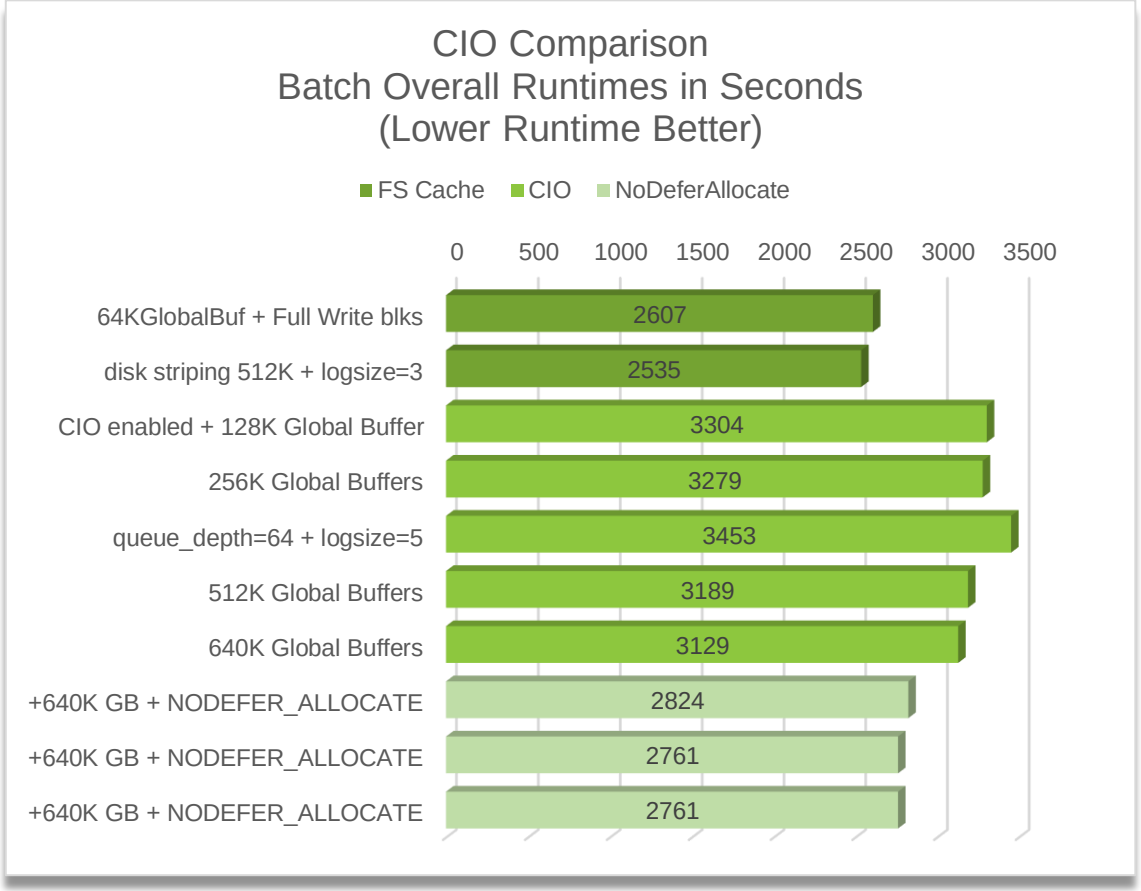
Recommended System Tunables

- **Profile gtmenv1 additions (Case sensitive)**
  - export LDR_CNTRL=DATAPSIZE=64K@STACKPSIZE=64K@TEXTPSIZE=64K@SHMPSIZE=64K
- **LPAR Changes (VIO LPAR and the Profile LPAR)**
  - chdev -l ent0 -a buf_mode=max -P
- **IOO Tuning**
  - ioo -p -o j2_nPagesPerRBNACluster=512
- **Queue depth for all application disks (gbls/journals)**
  - chdev -l hdiskX -a queue_depth=64 -U
- **Disk Tuning (For all application disks)**
  - # chdev -l hdiskX -a reserve_policy=no_reserve -a algorithm=shortest_queue -U
- **File System Settings**
  - # chfs -a logsize=5 /FILESYSTEM
  - rbw,rbr options for journal file systems (/profile_journals*)
  - log=INLINE -a options=noatime for database file systems (/profile_gbls*)
- **JFS2 Logical Volume Settings**
  - Inter=Policy = maximum, Stripe Size = 512K, Sched Policy = striped

CIO/ASYNC_IO Findings

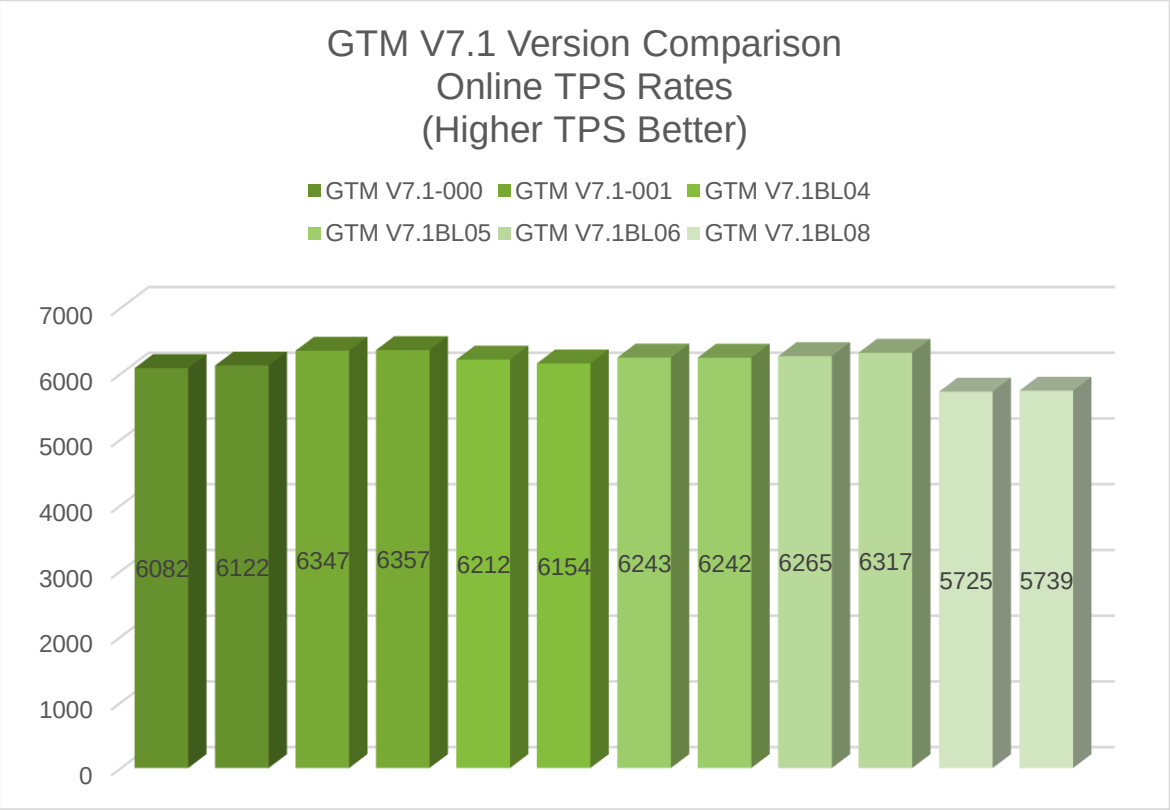
- **No significant improvement by enabling CIO on filesystem and ASYNCIO on database**
- **Use of File System Cache remains best performance solution**

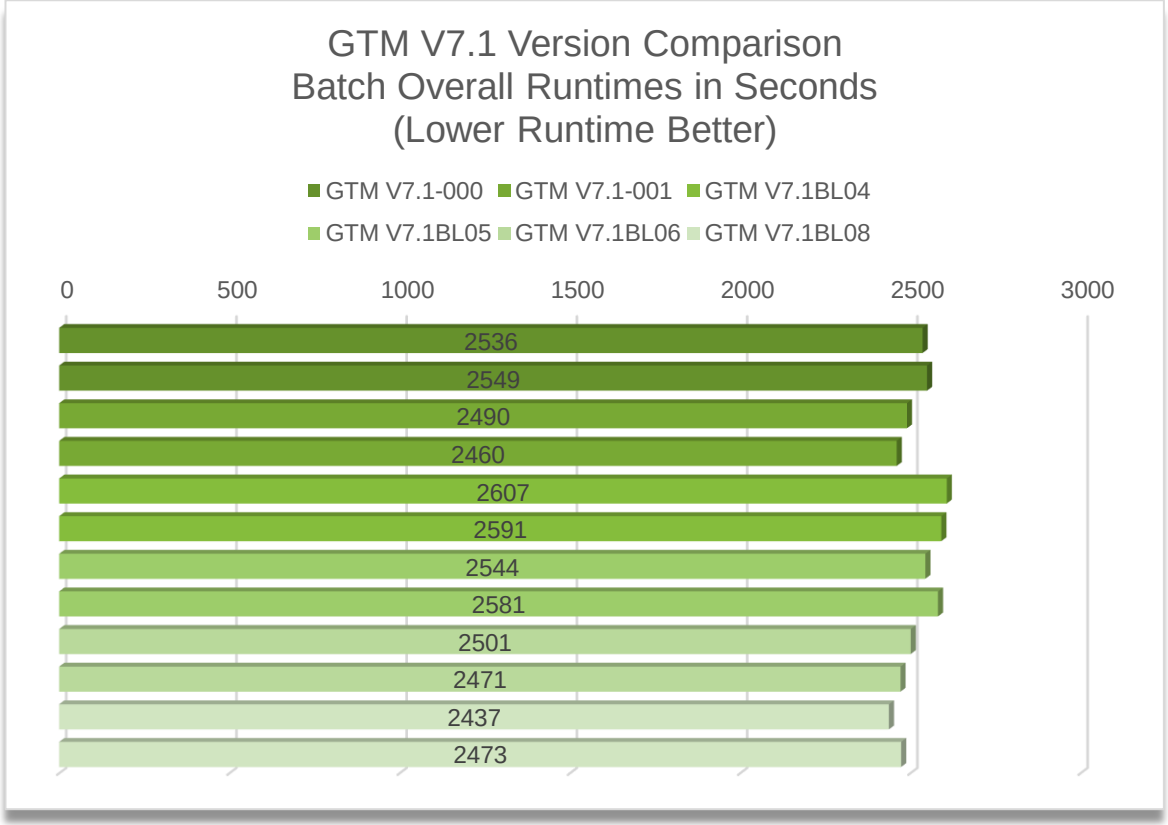




GTM Build Comparison Findings

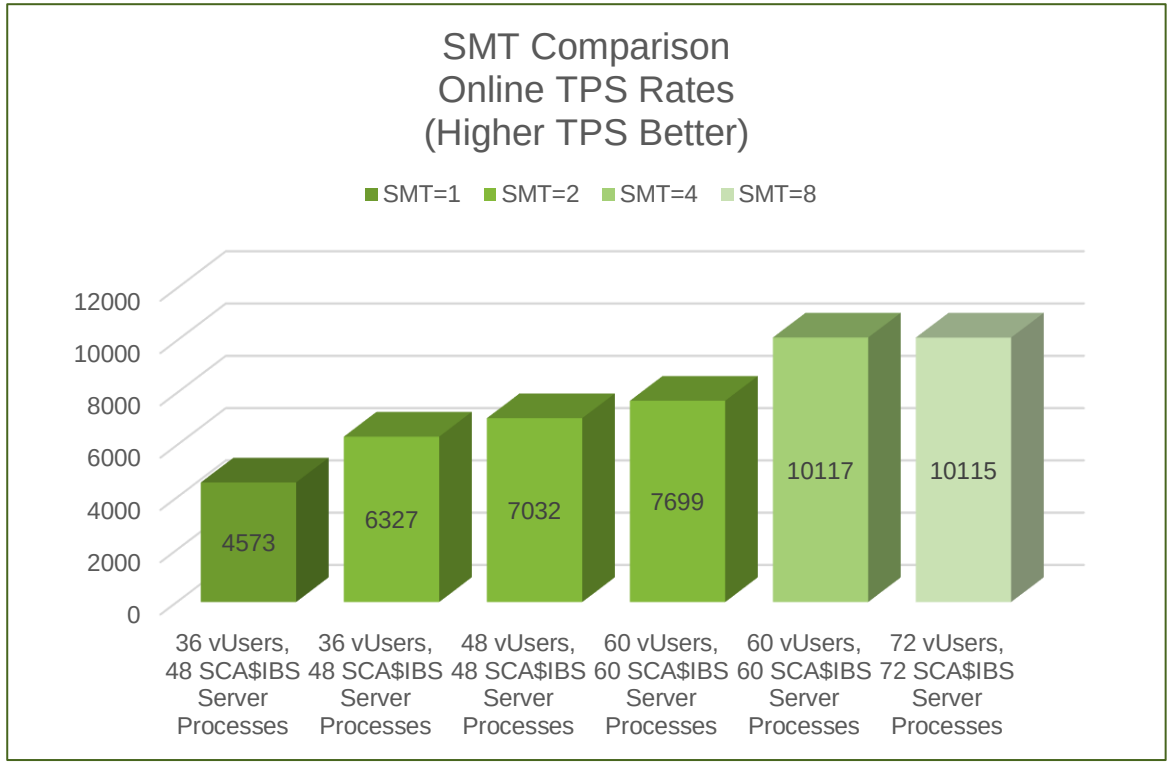
- **GT.M V7.1-001 Showed Best Overall Performance**
  - Individual test builds did not show any significant performance improvement

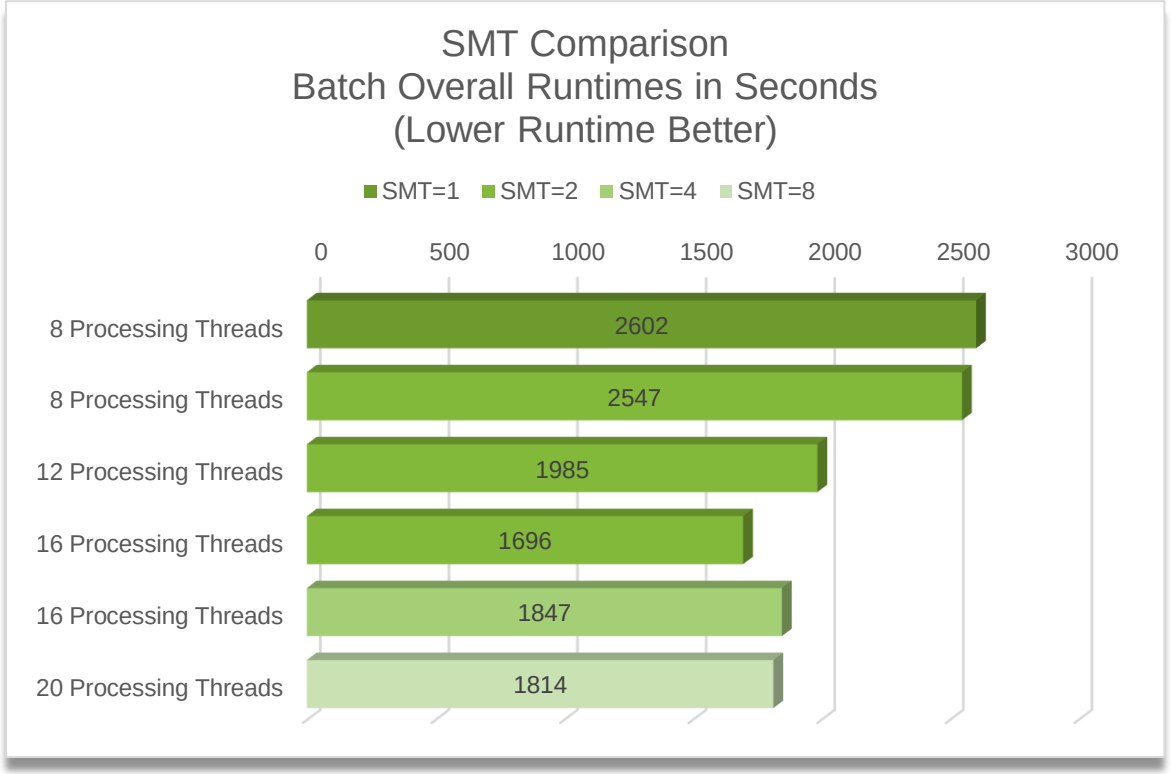




SMT Setting Findings

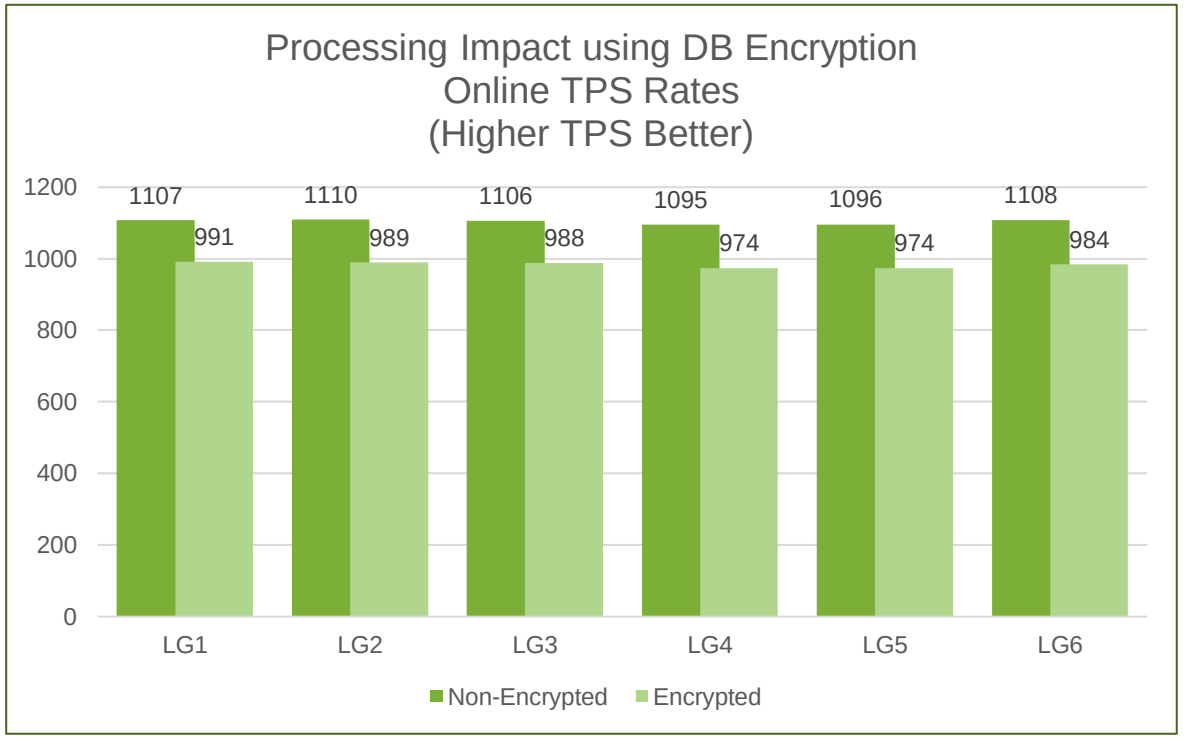
- **Latest SMT setting are best and final results**
  - As SMT value increased, additional online vUsers and batch processing threads were enabled

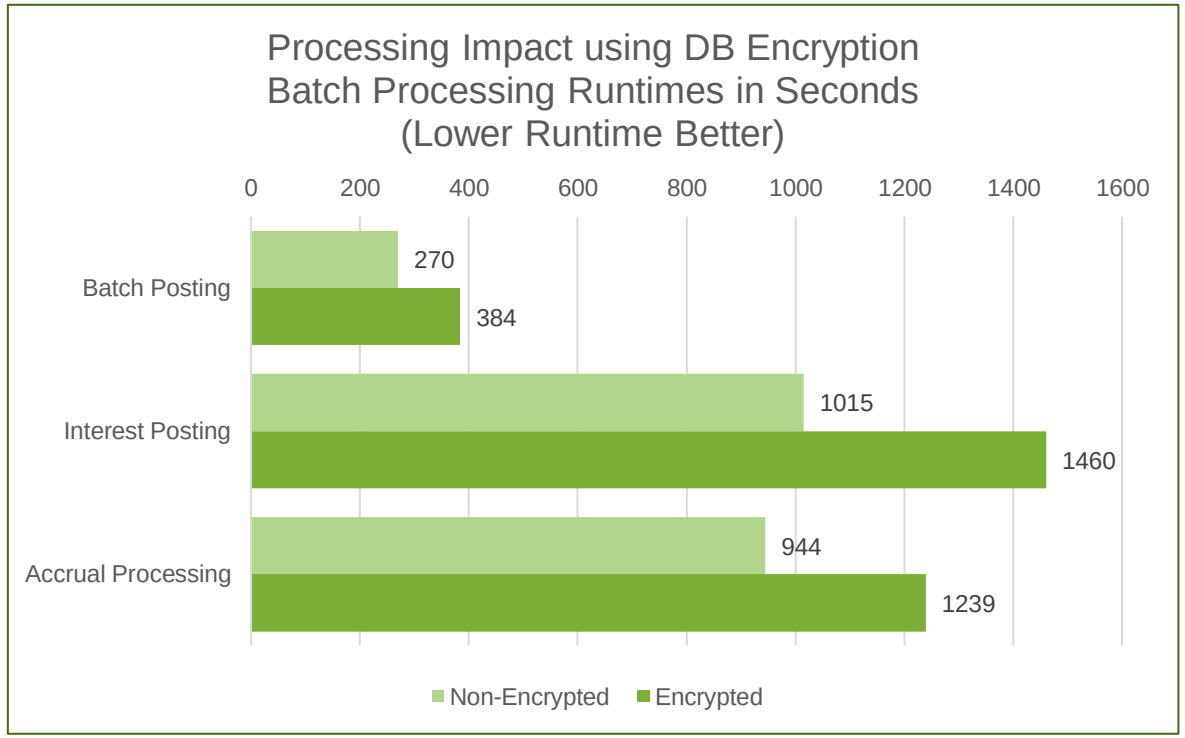




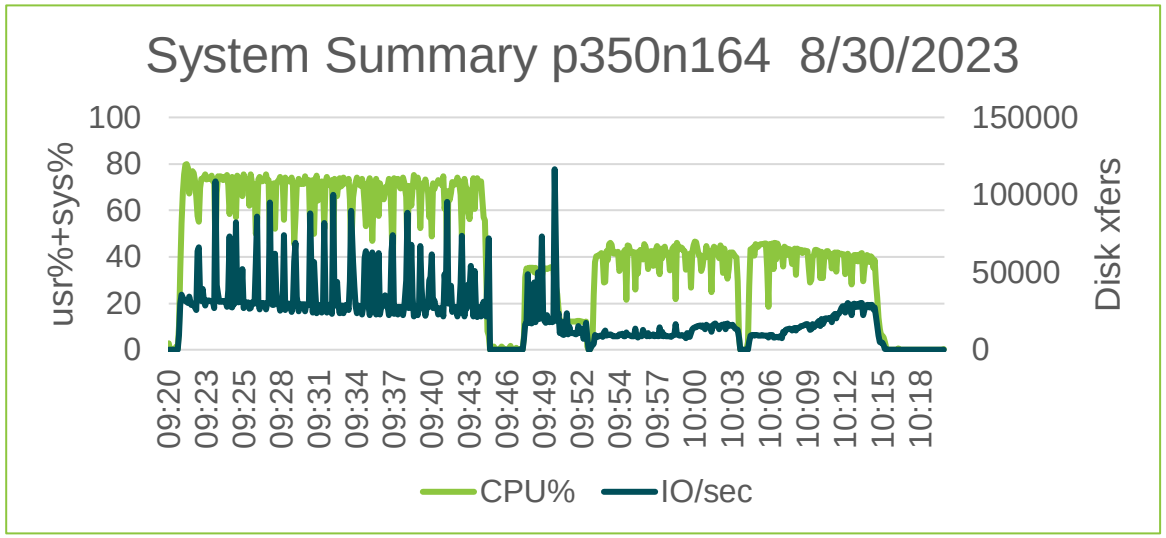
Profile/GT.M Database Encryption Findings

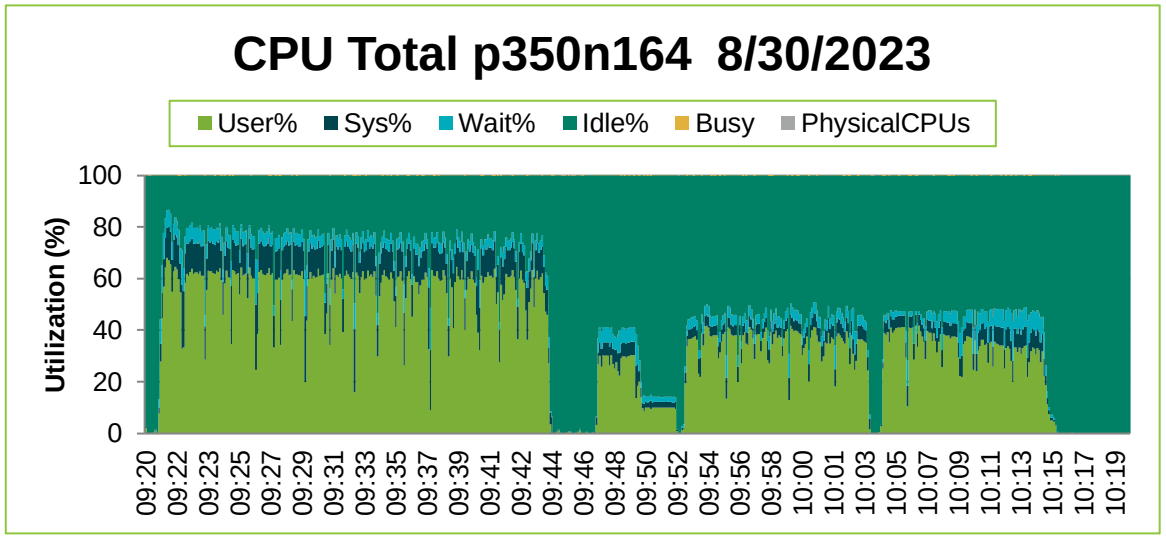
- Encryption Performance Impact (AES 256 Plug-in Library Architecture)
  - 11% Average Online Degradation
  - 28% Average Batch Degradation

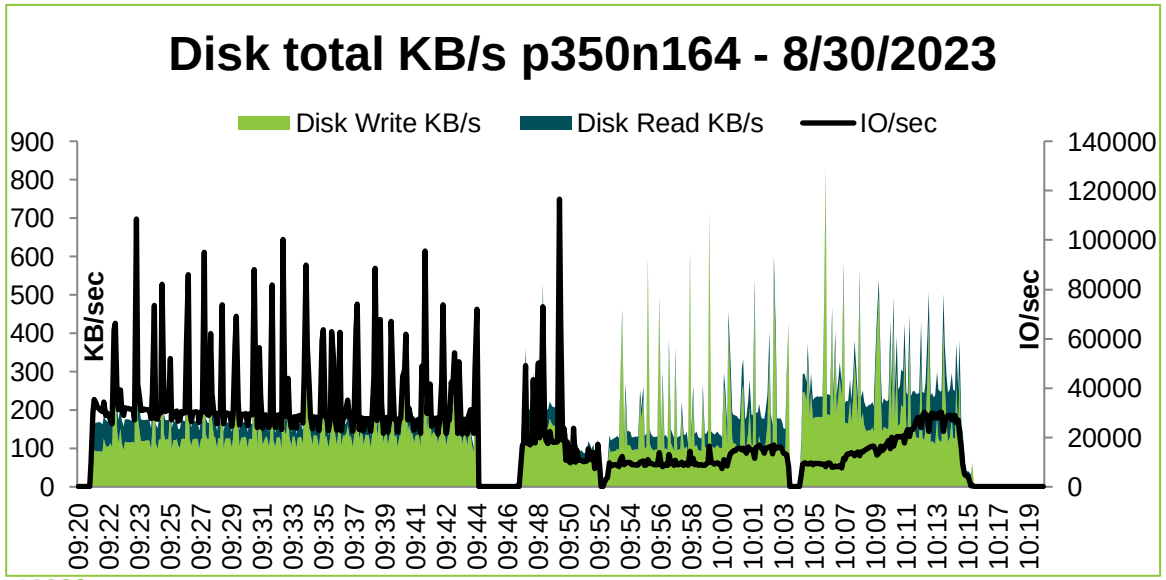


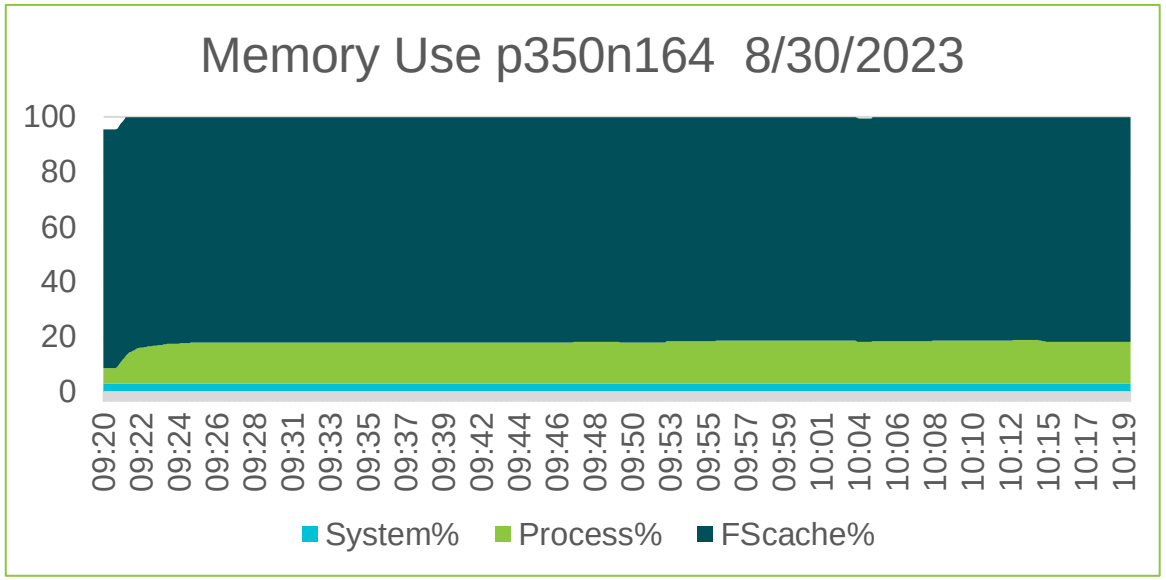


Host Server Utilization Statistics (SMT=2)









Next Steps / Future Performance Testing

- **Complete Documentation**
  - Produce Performance Summary Findings
  - Complete Internal Checklist (POWER10/AIX7.2)
- **Continued Performance Testing**
  - AES Encryption Optimization using POWER10 Accelerators
    - FIS Project – F246907
  - AIX7.3 Testing
  - Scaling to Larger Databases
  - GT.M POWER10 Internal Regression Testing